SMART SHOPPING CART WITH

AUTOMATIC BILLING SYSTEM USING ARDUINO

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Abstract:

Now a days, shopping has become a daily activity in today's world. We can see large queues in many shopping malls waiting for billing. The objective of our project is to overcome the problem of standing in queue and wasting time. To overcome the above problem, we are proposing a smart trolley billing system that will audit the purchased products and the payment is made online automatically using the RFID tag. It will automatically identify and scan the product, and the final billing is made from the cart itself. So that customers are free from waiting in a long queue at checkout. It also provides the centralized and automated billing system using RFID. This model is reasonable and profitable smart shopping cart handled by the IOT innovations. The primary goal is to provide a technology oriented, time saving and commercial oriented system for enhanced shopping experience. This system will also provide suggestions for the products based on user purchased history from a consolidate system. In this system, every product in mart will have RFID tag, and every cart will be having RFID Reader attached to it. These features will save time and make shopping easier. Overall, we can gain the best shopping experience.

In this project we will build a Smart Shopping Cart with an Automatic Billing System using EM-18 RFID Module & Arduino. The project is super easy to build and requires only a few electronics components. In this system, every product in Mart will have an RFID tag, and every cart will be having RFID Reader**.** When a product is scanned, the price is automatically added to the billing system. The LCD Display in this project shows the item name along with the price.

A supermarket is a place where customers come to purchase their daily using products and pay for them. So, there is a need to calculate how many products are sold and generate the bill for the customer. When we go to a shopping cart for shopping, we must work for selecting the right product. Also, after that, it is hectic to stand in line for billing all the goods. Hence, we are proposing to develop a smart shopping cart system that will keep the track of purchased products for billing using RFID and Arduino.

Index:

S.NO	Topic	Page. No
1.	Introduction	4
2.	Background	4
3.	Problem Definition	5
4.	Objectives	6
5.	Components used	7
6.	Methodology	8
7.	Working	9
8.	Results and discussion	10
9.	Conclusion and Future scope	11
10.	References	12
11.	Codes in Appendix	13

1. Introduction:

Shopping is simple but waiting on a bill counter makes shopping too boring and a tedious task. Huge amount of rush plus cashier preparing the bill with barcode scanner is too time consuming and results in long ques. This innovative project consists of an automated billing system which can be placed within the shopping trolley. This automated payment system consists of a RFID reader which is controlled by Arduino. So, whenever the shopper puts any product in trolley it is detected by the RFID module and is displayed on LCD along with the price of the product. As the shopper goes on adding products, all products are detected by the module and therefore the price will increase accordingly. In case if customer changes his/her mind and doesn't want any product added in the trolley he/she can remove it and the price added will be deducted automatically. At the end of shopping the shopper will press the button which when pressed adds all the product along with their price and gives the total amount to be paid. At exit for verification the shopkeeper can verify the products purchased with the help of master card. Hence this technique is an appropriate method to be used in places like supermarkets, this will help in reducing manpower and helps in making a better shopping experience for customers.

Background:

Individuals have constantly created innovation to bolster their requirements as from the start of humankind. The fundamental reason for development in innovation is ought for more independency and this leads to improving tasks and making regular one simpler and speedier. One significant task that individuals invest maximum measure of energy is in shopping. Shopping center is a spot where individuals get their everyday necessities running from sustenance items, garments, electrical machines and so forth. Some of the time clients have issues with respect to the unspecific data about the item marked down and misuse of superfluous time at the counters. In this innovative world, each grocery store and supermarkets utilize shopping trolleys with a specific end goal to help clients to choose and store the items which they expect to buy. Customers usually purchase the products required and place them in their carts and thereafter wait at the counters for payments of bills. The payment of bills at the counters is really troublesome and time-consuming process which thereby increasing a heavy crowd at the counters.

As indicated by a study directed by US Department agency, on a normal, people spend through 1.4 hours consistently on shopping. A considerable number of clients will tend to leave a line if the line is too long. The present Shopping environment can be essentially be characterized into two classifications

(1) Shopping in-individual

(2) Shopping in absentia.

Shopping in absentia is upheld from various perspectives including web shopping, online shopping, and so forth which will not require the buyer to be manually held at the Counters. Purchasing in-individual includes an individual call at location of purchasing and choosing items in view of different variables including need, comfort, brand, and so on. The proposed keen Shopping basket framework plans to help shopping in-individual that will minimize the time spent in shopping. Persistent change is required in the customary time spent at the counters to enhance the nature of shopping background to the clients. To beat these issues expressed above and to enhance the current framework, we have composed a Shopping basket. This can be done possible by basically connecting RFID labels to the items and reader with an LCD in the purchasing cart. From this framework client can possess data related to cost of each thing which are inside cart and furthermore absolute cost of the thing about the item. This framework will save time of clients and labor required in shopping Centre with respect to cost of the item.

Problem Definition:

In realistic, markets are these days utilized by a considerable number of individuals in order for securing most of items. Item procurement speaks to an unpredictable procedure that involves time spent in passageways, item area and checkout lines. Consumers commonly encounter some problems and difficulty during purchasing. These problems comprise worrying about the money which they have brought would be insufficient for all the items purchased and also dissipating a lot of time at the cashier. And also, it is becoming a increasing problem for the merchants to make their shoppers consigned and to anticipate 3 their demands because of the effect of contention and also because of lack of equipment that isolate application designs. At some instances clients have issues with respect to the inadequate data about the item of discounts and thereby misuse of superfluous time at the counters. We can end this issue by supplanting the omnipresent Universal Product Code (UPC) standardized identification by

keen names known as radio frequency identification (RFID) tag. To overcome the above problems, we implement the extensive notion of RFID based keen shopping cart in the field of retail stock.

Existing System:

In the existing system, they have used the traditional method of barcode scanning. Using the barcode scanner, we need to scan each product and so this method becomes very slow to be scanned. A barcode reader is associate in electronic device for reading with the barcodes. In this process we have no automatic billing system and the customer has wait for the billing process in the long queues. Therefore, using the barcode process billing method is slow. This eventually results in the long queues. To avoid the process, we introduced types of technology is the RFID based billing system. User can pay the amount through credit/debit cards or by cash. But it is the time consumption process for the billing purpose. So, the waiting time to pay the bill is increased. To overcome, the time consumption process the RFID based smart trolley system is proposed.

Objectives:

The main objective involved in this plan is to implement a smart shopping cart with the help of RFID technology for improvising purchasing. The plan is to employ the RFID related surveillance implementation practice in the purchasing cart. In this plan RFID card is utilized as protection entry for acquiring of commodities in the shopping malls. If the commodity has been placed in the shopping cart the price of the product appears and accordingly the total amount will be shown and if we wish to remove the product from the trolley, you can take away the product and the amount of that specific product gets deducted from total amount. In this, the technology used is for obtaining the products thereby which boosts security performance and speed while purchasing in shopping complexes. The technological objective for our presented problem in shopping complexes is the practice of RFID technology for the instinctive recognition of commodity in the interior of the purchasing cart thereby annihilating shopper intervening in the task of commodity purchase and for payment. The principal point of proposed framework is to give an innovation which is minimal effort oriented, effectively adaptable, and efficiently feasible for helping shopping in individual. With the help of this a lot of time will be saved at the billing counters.

Components used:

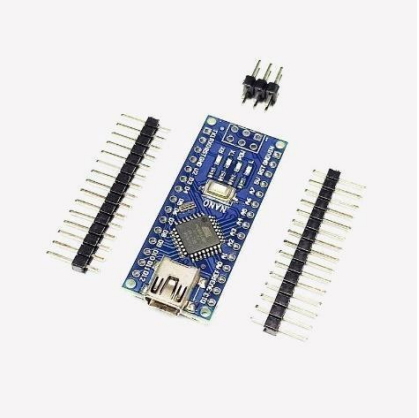


Fig.1. Arduino Nano Board

Fig.2. RFID Reader EM-18

Fig.3. Push Button Switch

Fig.4. LCD Display

Fig.5. RFID Card

Fig.6. Green LED and Red LED 5mm

Fig.7. Potentiometer 10K

Fig.8. Battery

Methodology:

The main controller of this project is Arduino Nano Board where the project code is stored. The Arduino operates all the processes from reading RFID Cards info to storing and calculating prices. The RFID Module that we are using here is the EM-18 Module which operates at a frequency of 125KHz. Apart from the RFID Cards, we need 5-10 RFID cards. These RFID cards are named as different items name like tea, milk, biscuits, oil, etc.
The system has a buzzer that turns on when an RFID is scanned. There are two LEDs Red and Green LED. These LEDs turn on and off based on items added or removed. The most important part of the project is the LCD Display. We are using a 16×2 LCD display for this project. The 10K potentiometer is used with LCD Display for adjusting LCD contrast. When you press hold the reset button and scan the RFID, the items can be removed.

To power the entire circuit, we will use a pair of Lithium-Ion Batteries. Each 3.7V Samsung 18650 connects in series to give a 7.4V Supply. Then the 7.4V is supplied to Vin Pin of Arduino Board. The RFID and LCD module is powered via a 5V pin of Arduino Nano.

The Buzzer, Green LED and Red LED connects to digital pins 6, 5, and 4 of Arduino. The RFID Module EM-18 Tx pin connects to the RX0 pin of Arduino. The Reset button connects to the A4 Pin of Arduino. Pin 4, 6, 11, 12, 13, 14 of LCD connects to Pin 12, 11, 10, 9, 8, and 7 pins of Arduino.

Working:

1. After uploading the code, you can start testing the device. Initially, the LCD display will show the message “Welcome to Super Market”.

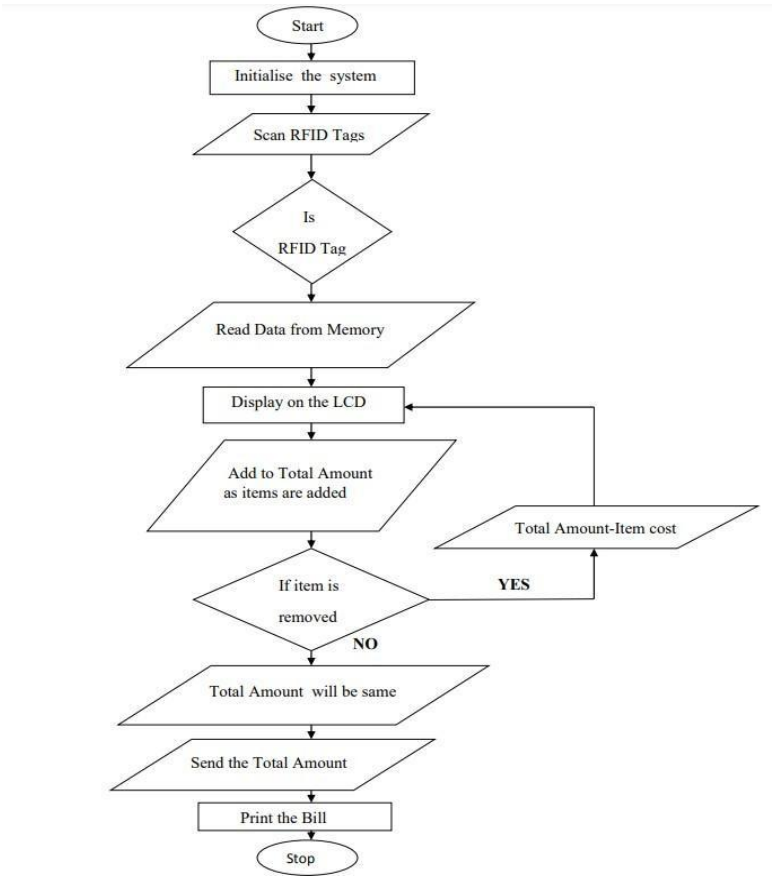
2. The LCD will display the message to Add the item. You can add any item simply by scanning the item with an RFID tag.

3. For example, the RFID Card marked as Tea is scanned in our project. The LCD will display Tea is Added along with the price value of the tea.

4. You can scan other cards with different item names like Milk, Butter, or whatever you have assigned in your code. Every time you add any item, the price is added to the previous value with a total cost.

5. In case you want to remove any item from the list, then press and hold the reset button and scan the item. The item will be removed from the system and the price will be updated automatically. At the end of shopping the shopper can show the total price value of the products and pay the final price.

Flow chart of procedure:



Results and discussion:

This way the system reduces the physical task of the customer load effectively. RFID is inexpensive, average processing time of RFID is 0.99. The RFID having status updating of complete listings of the products purchased, automatic bill generation, updating stock inventory management to the central system.

No more long queues of customers waiting to pay. The customers just walk out smoothly. Customers return to your store 114% more since no checkouts means a stress free shopping experience! Reduce the number of hours staff wait at a register, and let them do the job only people can do, like attending your customers.

RFID tags don't need to be positioned in a line of sight with the scanner. RFID tags can be read at a faster rate than barcodes; as approximately 40 RFID tags can be read at the same time. RFID tags can work within much greater distances; information can be read from a tag at up to 300 ft.

Initially, this might be an expensive method as RFID tags are costly. But once started and being used, they reduce long term expenses and are helpful.

Fig.9. Components before Implementation

Fig. 10. Result after adding one product

Fig.11. Result after adding two products

Conclusion and Future Scope:

In spite of the fact that the RFID can't totally supplant the standardized identification innovation because of higher cost precision and speed is high in RFID implementation. The retailers, makers and purchaser products organizations like CVS, Tesco, Prada, Benetton, Wal-Shop and Procter and Bet are presently actualizing the innovation and investigating the effect of the innovation on their business. Others can likewise implement RFID. The basis of accomplishment lies in understanding the innovation and different components to minimize 4 the potential issues. It is time for the industry ought to begin using the new innovation, for example, RFID in different applications, for example, fabricating, and storing purposes. In this project the items in the cart is restricted to less than 50 as the customers don't prefer excess luggage at once. Tags used here can be attached to the products only in the longitudinal fashion without any folds. Tags given for the reader are of length 8cm and of width 2cm and they need to be attached to the products in a visible fashion for the reader then only the reader can detect tags without any fail. Tags used in this project are water sensitive and metal sensitive and have the capacity of reading only one side. So, the trolley is now restricted to use water sensitive packaged and metal sensitive packaged products. But the problem can be rectified with the waterproof tags and metal resistant tags which are under research at present. And the multiple object RFID reader used here has the constraint of skipping the tags at sometimes. Reader which can detect less than 50tags/sec is employed according to the requirement and meanwhile cost will also be under control for implementation of system.

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Appendix

Sample code of my Project:

Libraries included:

```
#include <Wire.h>
#include <SPI.h>
#include <MFRC522.h>
#include <LiquidCrystal_I2C.h>
```

Declaring Variables:

```
int pr1=0,pr2=0,pr3=0,pr4=0,pr5=0,pr6=0,tot=0,cnt;
int count=0,c1=0,c2=0,c3=0,c4=0,c5=0,c6=0;
```

String to store Tag UID:

```
String tagUID = "89 50 E5 7C";
 String tagUID1 = "79 3F 1D 85";
 String tagUID2 = "A7 55 CA 4D";
 String tagUID3 = "F7 E4 40 4E";
 String tagUID4 = "A7 9C 1D B3";
 String tagUID5 = "0A 56 11 67";
```

To Read the RFID Card:

```
String tag = "";
 for (byte j = 0; j < mfrc522.uid.size; j++)
 {
    tag.concat(String(mfrc522.uid.uidByte[j] < 0x10 ? " 0" : " "));
    tag.concat(String(mfrc522.uid.uidByte[j], HEX));
```

To add/remove a product:

```
if (tag.substring(1) == tagUID)
  {
    pr1++;
    digitalWrite(buz,1);
//    Serial.print("Prod-1 of Rs-25");
     lcd.setCursor(0,0);
     lcd.print("prod-1 of Rs-25 ");
     digitalWrite(buz,0);

  if(pr1%2==1)
   {
    digitalWrite(buz,1);
    digitalWrite(gled,1);
    delay(100);
    digitalWrite(gled,0);
    digitalWrite(buz,0);
    tot=tot+25;
    cnt=cnt+1;
   }
  else
  { digitalWrite(buz,1);
   digitalWrite(rled,1);
   delay(100);
   digitalWrite(rled,0);
   digitalWrite(buz,0);
   tot=tot-25;
   cnt=cnt-1;
  }
   delay(2000);
   goto ss;
}
```